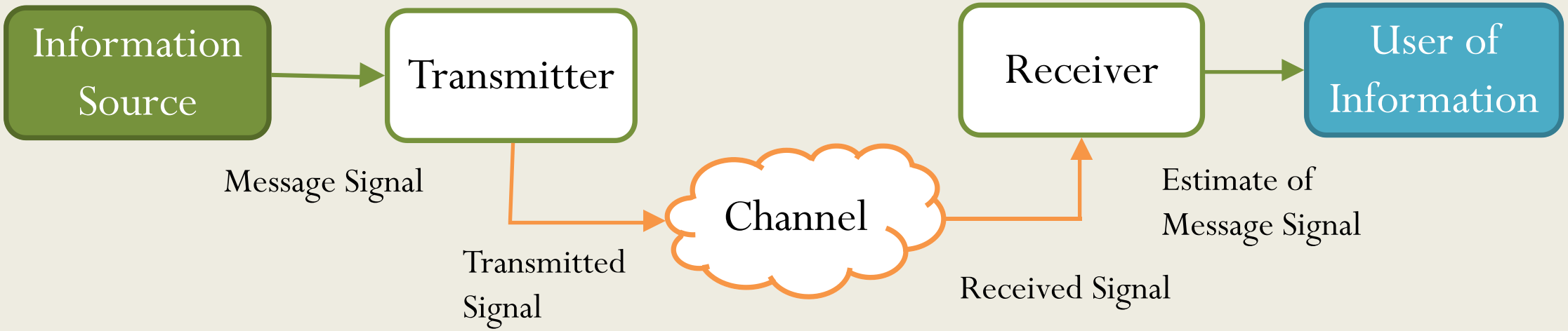
Encoding and Modulation

Encoding and Modulation

- Analog to Analog signal conversion – AM,FM, PM
- Analog to Digital signal conversion – ASK,FSK,PSK, QAM
- Digital to Analog signal conversion – PCM, DM
- Digital to Digital signal conversion – Line & Block Coding

Communication System

- Communication involves the transfer of information or intelligence from a source to a receiver via a channel or medium.
- Block Diagram of a Communication System



Basic Blocks

- ***Source:*** analog or digital Signal
- ***Transmitter:*** transducer, amplifier, modulator, oscillator, power amplifier, antenna
- ***Channel:*** e.g. cable, optical fiber, free space
- ***Receiver:*** antenna, amplifier, demodulator, oscillator, power amplifier, transducer
- ***User:*** e.g. Person, speaker, Computer

Information Source

- The source of information may be characterized in terms of the signal that carries the information.

Signal can be

- *One dimensional - speech, music or computer data.*
- *Two dimensional - pictures*
- *Three dimensional - video*
- *Four dimensional – volume data over time*

Transmitter

- The transmitter has a collection of Electronic Components and Circuits (Such as transducer, amplifier, modulator, oscillator, mixer, antenna, etc.,) designed to convert the message signal into a suitable form for propagating over the communication channel.
- The transmitter increases the power level of the signal
- It is increased in order to cover a large distance

Communication Channel (or) Medium

- The medium used for transmission of electronic signals from one place to the other.
- The communication medium can be conducting wires, coaxial cable, optical fibers (or) free space

E.g.

- twisted pair – telephone cable
- coaxial cable –Thick black cable used for higher bandwidth communications than twisted pair (i.e. Optus cable)
- fibre optic – data transferred through pulses of light. Extremely fast.
- Non cable methods (free Space) such as satellite, microwave, wireless and Bluetooth

Two types of Communication Channel

1. Wired Communication (or) Line Communication (or) Guided Propagation
2. Wireless Communication (or) Radio Communication (or) Free Propagation

Wired Communication

- Simple wires, coaxial cables , optical fibers.

Example

- Telegraph, telephone systems, cable TV etc.,
- Due to physical connection, these systems cannot be used for long distance communication

Primary communication resources

Two resources

- Transmitted power (Average power of transmitted signal)
- Channel bandwidth (band of frequencies allocated for transmission of the message signal)
- one resource may be considered more important than the other

Therefore the classification of communication channel as

- ***Power limited***
- ***band limited***

Noise or Distortion

- **Noise** – unavoidable presence in communication Systems.
- Unwanted waves that tend to disturb the transmission and processing of message signals in a communication Systems.
- Source of noise may be Internal or External to the system

Various types of Noise Sources

- Internal – Electronic Devices
- External – Man made noise

Natural Noise

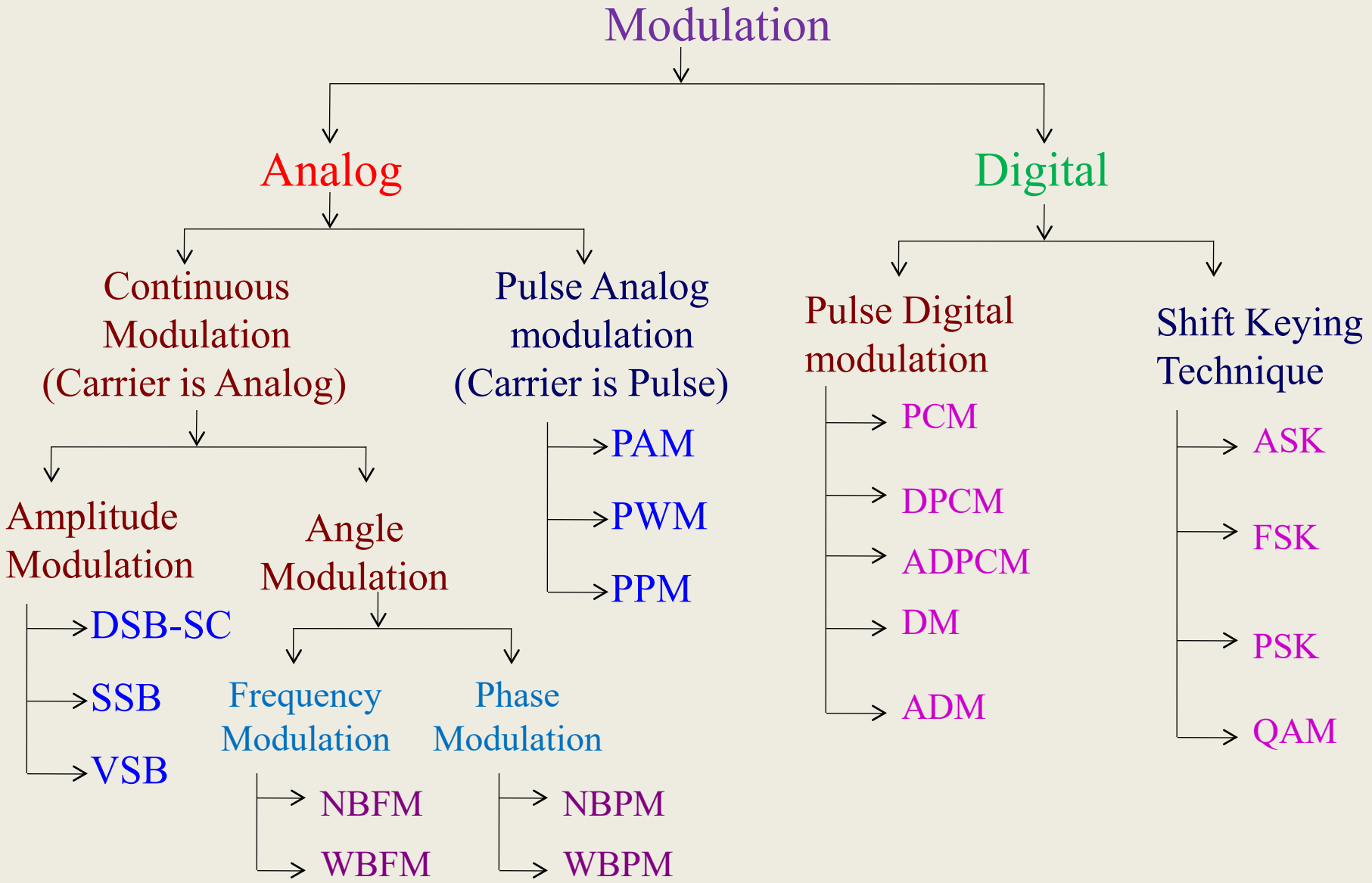
Receiver

- It consists of electronic circuits like antenna, amplifier, demodulator, transducer etc.,
- The reception is exactly the opposite process of transmission.
- Output transducer converts the electrical signal into original form (i.e) voice or TV picture etc.,

Recipient (or) Destination

- The destination is the person (or) loud speaker (or) computer monitor. It receives voice (or) music (or) computer data

Types


Modulation
Analog
Digital
Continuous Modulation (Carrier is Analog)
Pulse Analog modulation (Carrier is Pulse)
Pulse Digital modulation
Shift Keying Technique
Amplitude Modulation
Angle Modulation
PAM
PWM
PPM
PCM
DPCM
ADPCM
DM
ADM
ASK
FSK
PSK
QAM
DSB-SC
SSB
VSB
Frequency Modulation
Phase Modulation
NBFM
WBFM
NBPM
WBPM

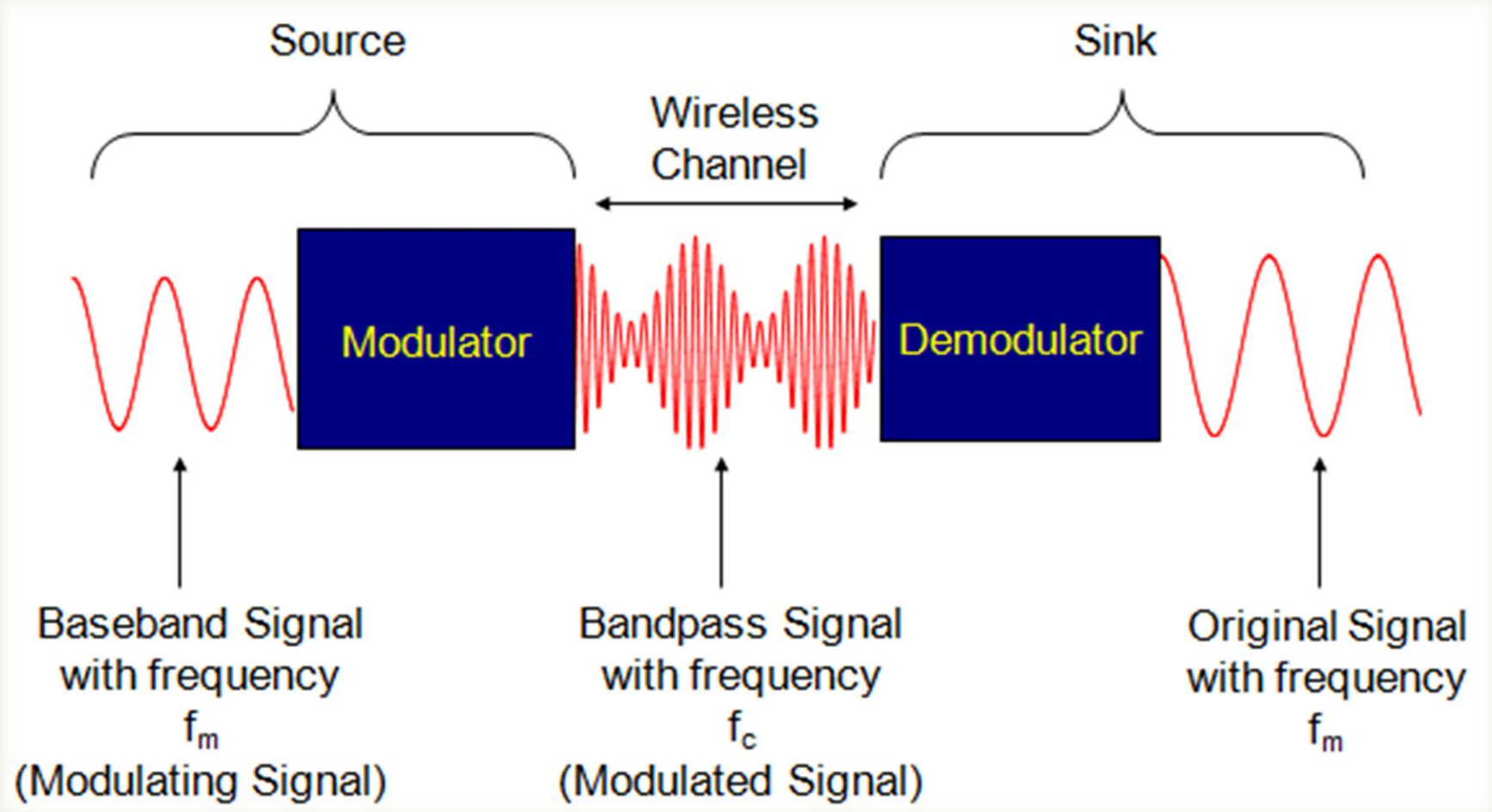
Modulation

- To reduce the size of the antenna required
- Long Distance Communication
- To prevent mutual interference between stations
- Avoids Mixing of Signals
- Reduces the effect of Noise to improve quality of Reception

Definition

- The process of Changing the characteristics of carrier signal or high frequency signal in accordance with the amplitude of modulating signal/ Information Signal/ low frequency message signal.

Modulation



Picture Taken from: https://padakuu.com/modulation-and-demodulation-32-article

Analog

- Amplitude Modulation
- Angle Modulation

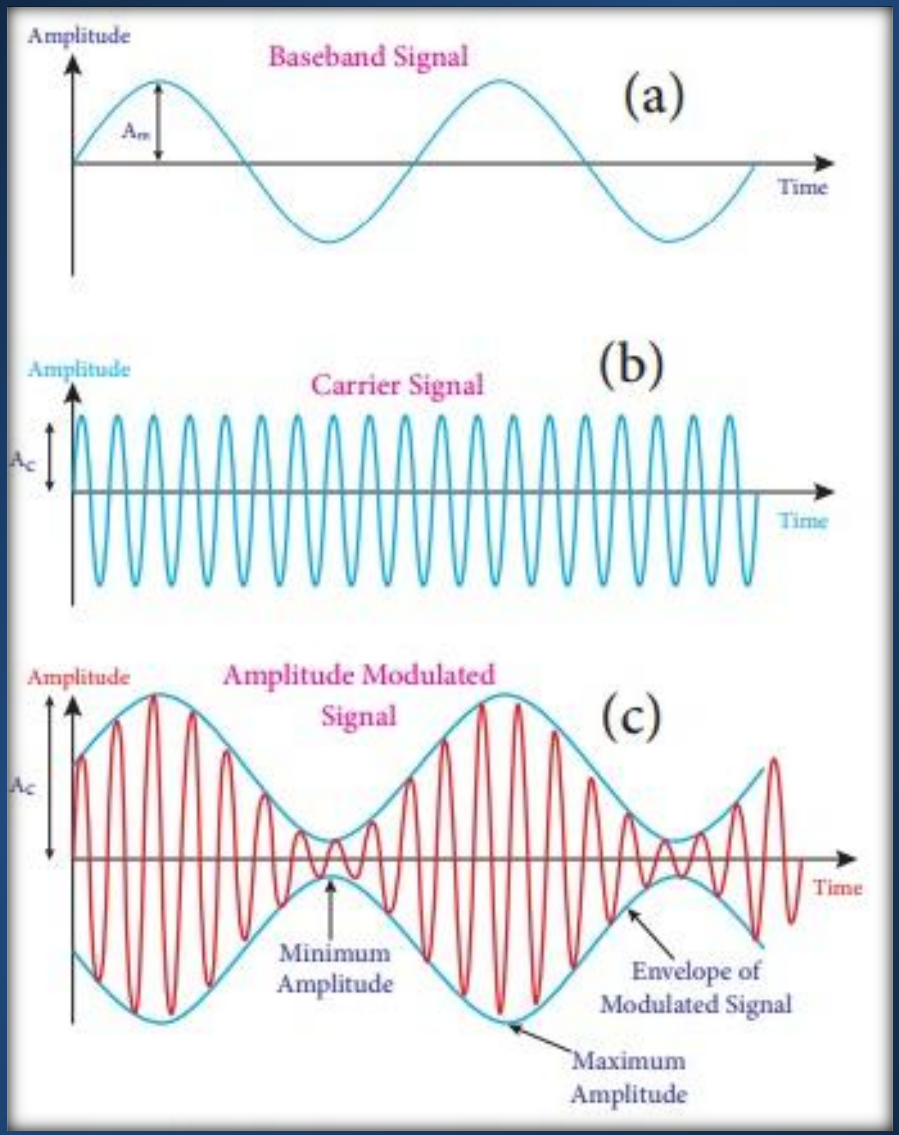
  Frequency Modulation

  Phase Modulation

The AM signal: $s(t) = A_c[1 + k \bullet m(t)]\cos \omega_c t$

The modulating signal: m(t)

The Carrier Signal: $c(t) = A_c \cos \omega_c t$



Picture Taken from: https://www.brainkart.com/article/Amplitude-Modulation-(AM)_41571/

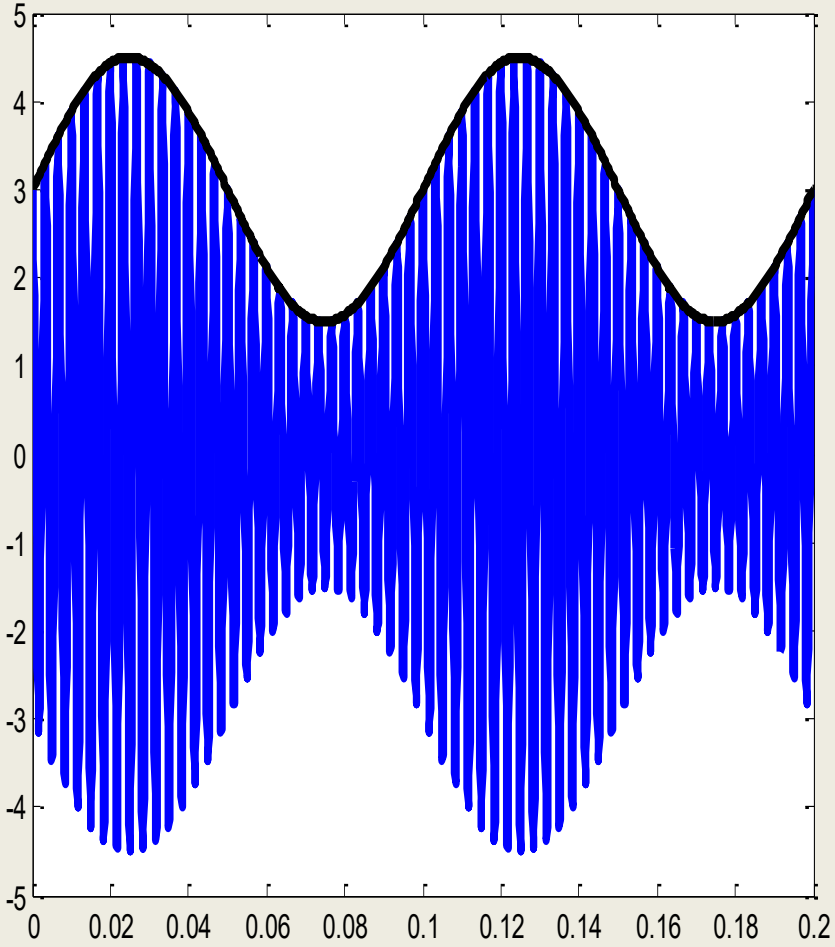
Analog

- AM Envelope

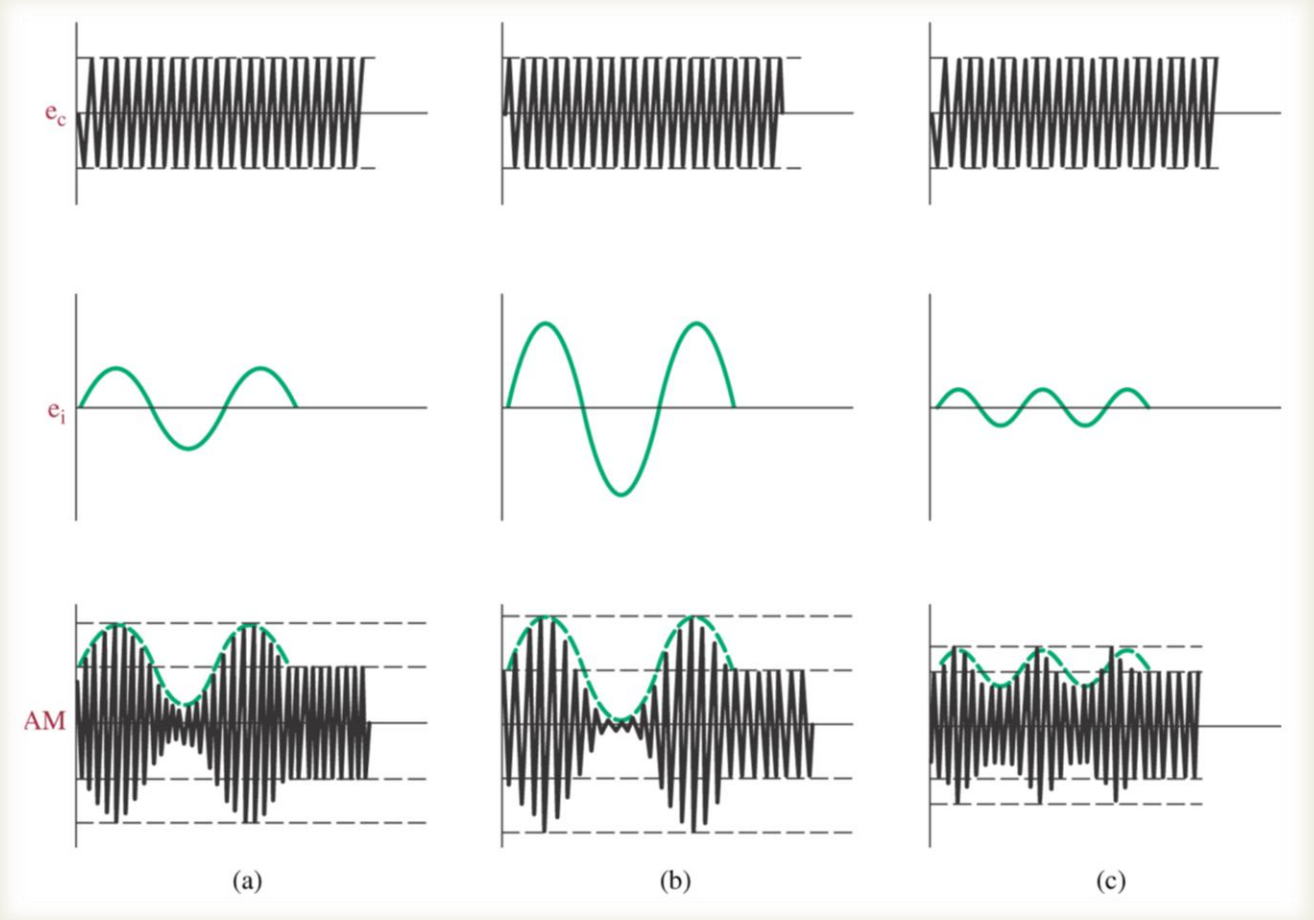
$$s(t) = A_c[1 + k \bullet m(t)]$$

- AM Signal

$$s(t) = A_c[1 + k \bullet m(t)]\cos \omega_c t$$



AM Envelope The shape of modulated signal is defined as AM envelope because it contains all frequencies that make up AM signal.



Representation of AM

Let $V_m(t)$ be the baseband signal and $V_c(t)$ be the carrier signal given by

$$V_m(t) = V_m \sin \omega_m t$$

$$V_c(t) = V_c \sin \omega_c t$$

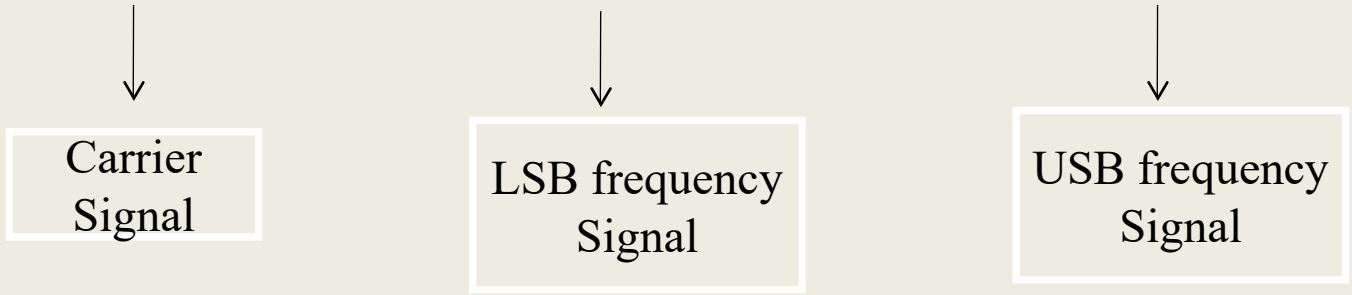
Amplitude of the carrier signal is changed after modulation.

$$V_{AM} = V_c[1 + m_a \sin \omega_m t] \sin \omega_c t$$

Where,

$m_a = V_m / V_c$ modulation Index (or) Depth of Modulation.

$$V_{AM} = V_c \sin \omega_c t + m_a V_c /2 \{\cos(\omega_c - \omega_m)t - \cos(\omega_c + \omega_m)t \}$$



AM Spectrum and Bandwidth

Bandwidth: The amount of data which can be transmitted on a medium over a fixed amount of time (second).

Sum of frequency = $f_c + f_m$

Difference in frequency = $f_c - f_m$

The AM spectrum extends from $f_c - f_m$ to $f_c + f_m$

Where f_c – carrier signal frequency

f_m – modulating signal frequency

Contd…

- The Band of frequency between $f_c - f_{m(max)}$ and f_c is lower side band(LSB). Any frequency within this band is called Lower side band frequency (LSF).
- The Band of frequency between f_c and $f_c + f_{m(max)}$ is upper side band(USB). Any frequency within this band is called upper side band frequency (USF).

The bandwidth of an AM wave is equal to the difference between Highest USF and Lowest LSF (or) 2 times the highest modulating signal frequency.

$$B = [f_c + f_{m(max)}] - [f_c - f_{m(max)}]$$
$$= 2 f_{m(max)}$$

Modulation Index

The amount of amplitude change (modulation) present in an AM waveform is co-efficient of modulation (or) Depth of modulation (or) modulation index.

$$m = E_m / E_c$$

E_m – peak change in the amplitude of the output waveform voltage

E_c - peak amplitude of the unmodulated carrier voltage

$$E_m = \tfrac{1}{2} [\, V_{max} - V_{min} \,]$$

$$E_c = \tfrac{1}{2} [\, V_{max} + V_{min} \,]$$

$$V_{max} = E_c + E_m$$

$$V_{min} = E_c - E_m$$

$$E_m = E_{USF} + E_{LSF} \quad \text{and } E_{USF} = E_{LSF} = E_m/2$$

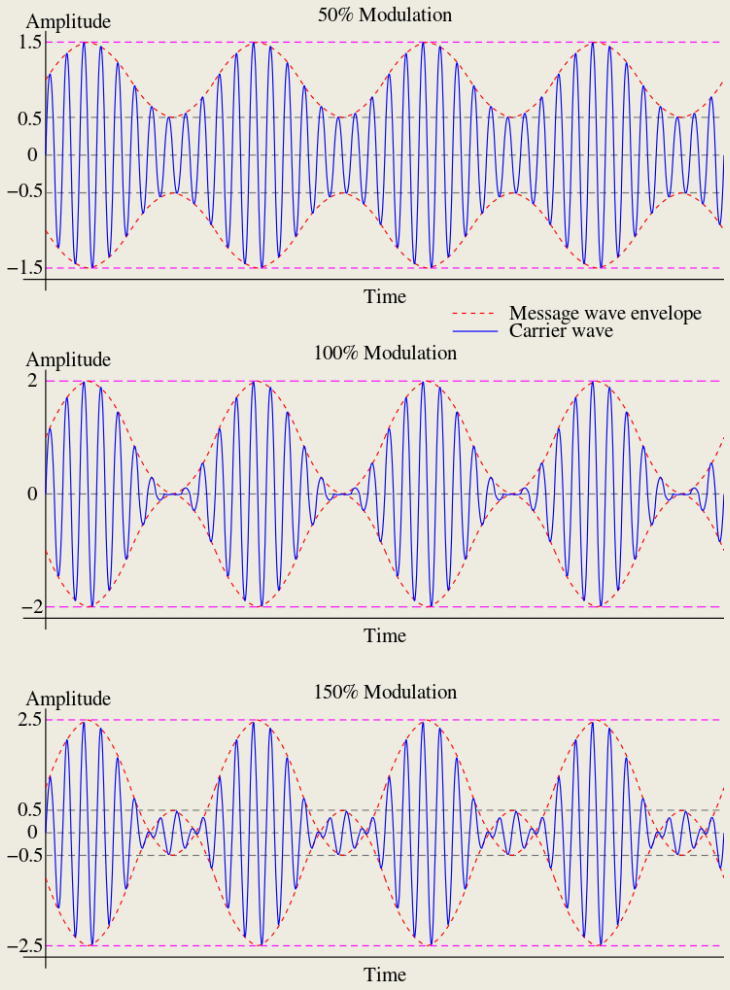
Percentage modulation

Percentage change in amplitude of the output wave when the carrier is acted by a modulating signal.

$$M = (E_m / E_c)*100 \text{ (or) } m*100$$

Classification based on modulation Index

- 100% (or)critical modulation, m =1
- 50% (or) under modulation, m<1
- Over modulation, m>1



Picture Taken from:
https://commons.wikimedia.org/wiki/File:Amplitude_Modulated_Wave-hm-64.svg

Frequency Modulation

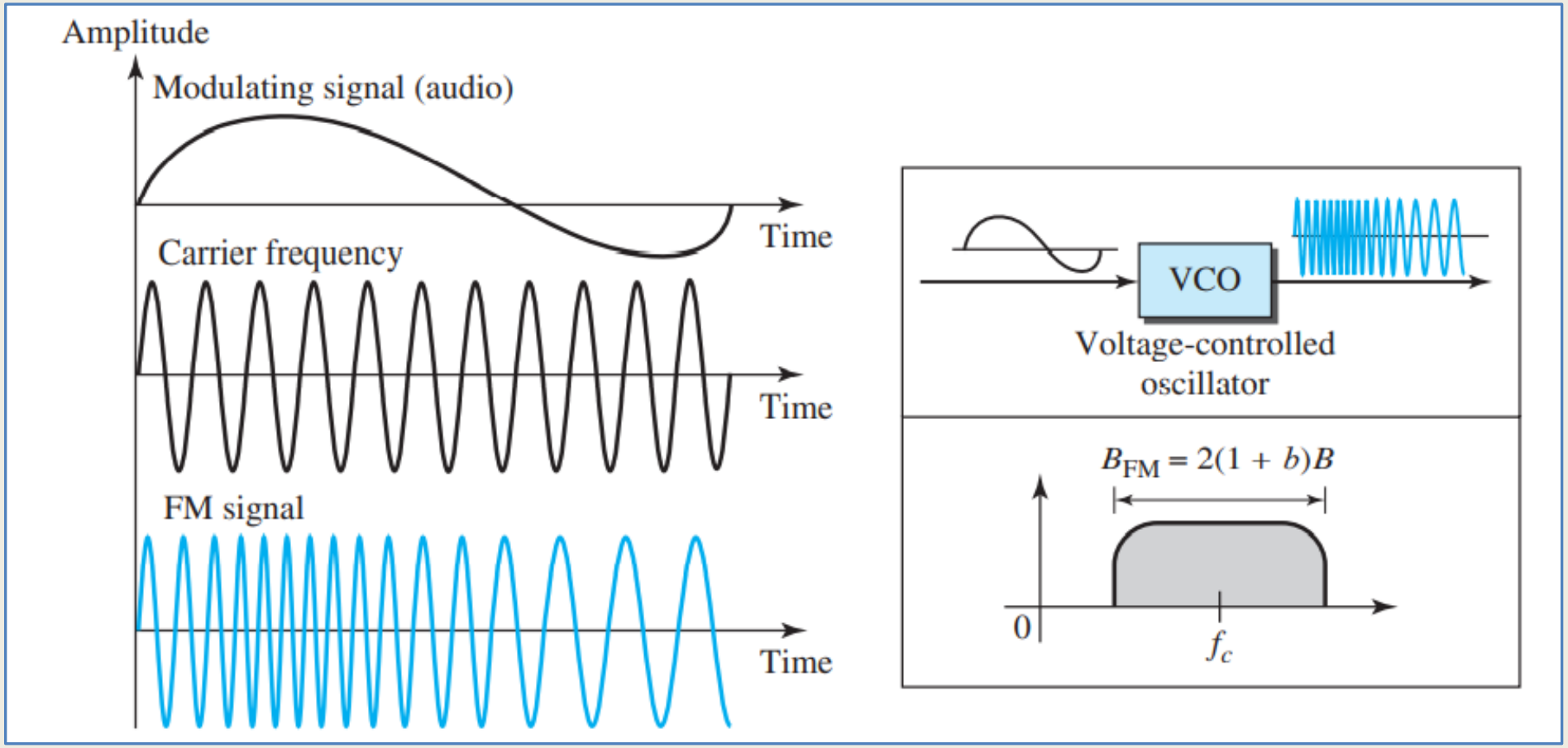
- In FM transmission, the frequency of the carrier signal is modulated to follow the changing voltage level (amplitude) of the modulating signal.
- The peak amplitude and phase of the carrier signal remain constant, but as the amplitude of the information signal changes, the frequency of the carrier changes correspondingly.

Definition

- The process of Changing the frequency of carrier signal or high frequency signal in accordance with the amplitude of modulating signal/ Information Signal/ low frequency message signal.

- The total bandwidth required for FM can be determined from the bandwidth of the audio signal: $B_{FM}=2(1\times ß)B$.

Contd...


Amplitude
Modulating signal (audio)
Time
Carrier frequency
Time
FM signal
Time
VCO
Voltage-controlled
oscillator
$B_{FM} = 2(1 + b)B$
0
f_c

Phase Modulation

- In PM transmission, the phase of the carrier signal is modulated to follow the changing voltage level (amplitude) of the modulating signal.
- The peak amplitude and frequency of the carrier signal remain constant, but as the amplitude of the information signal changes, the phase of the carrier changes correspondingly.

Definition

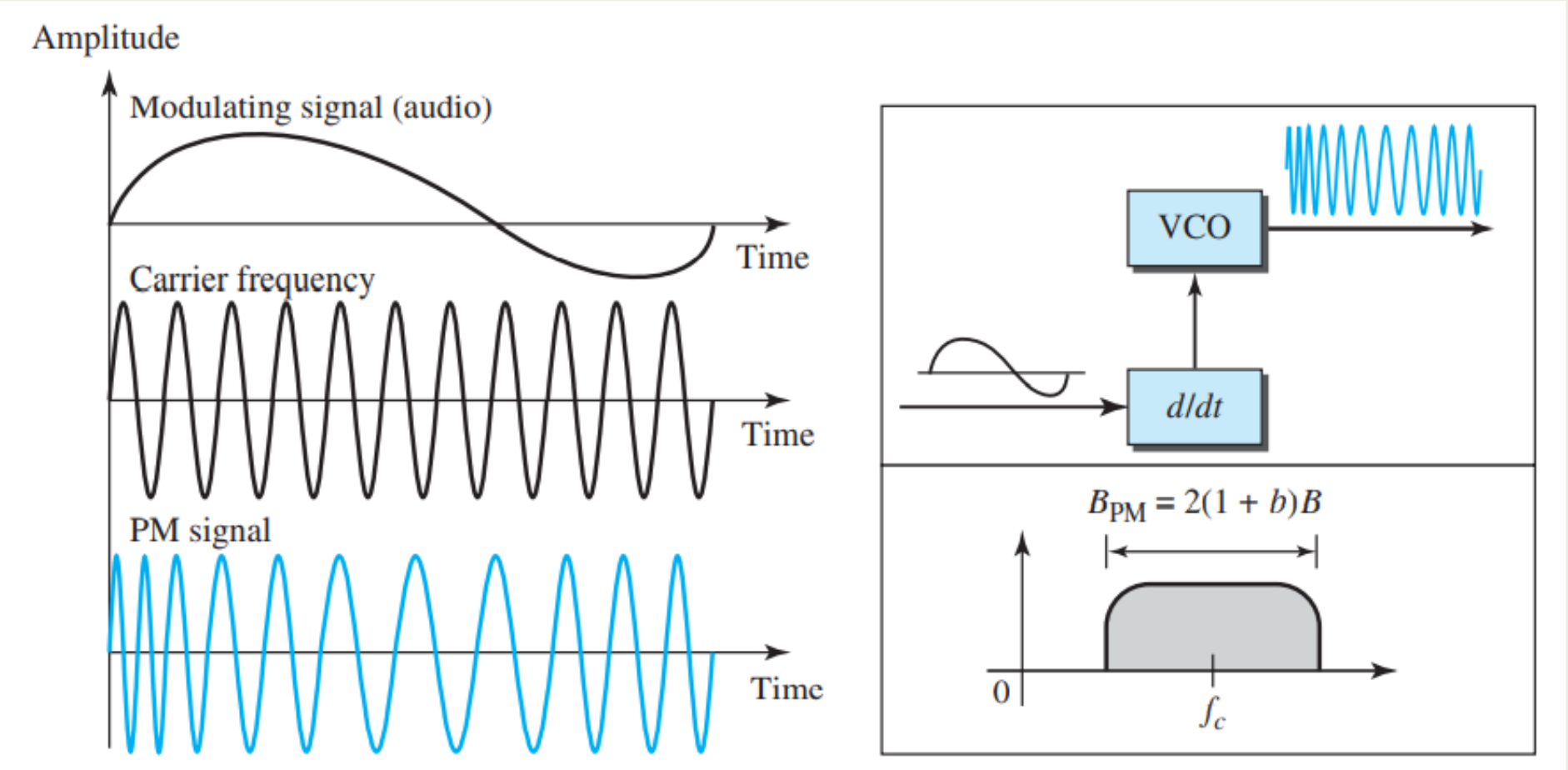
- The process of Changing the Phase of carrier signal or high frequency signal in accordance with the amplitude of modulating signal/ Information Signal/ low frequency message signal.

- The total bandwidth required for PM can be determined from the bandwidth and maximum amplitude of the modulating signal: $B_{PM}=2(1\times ß)B$.

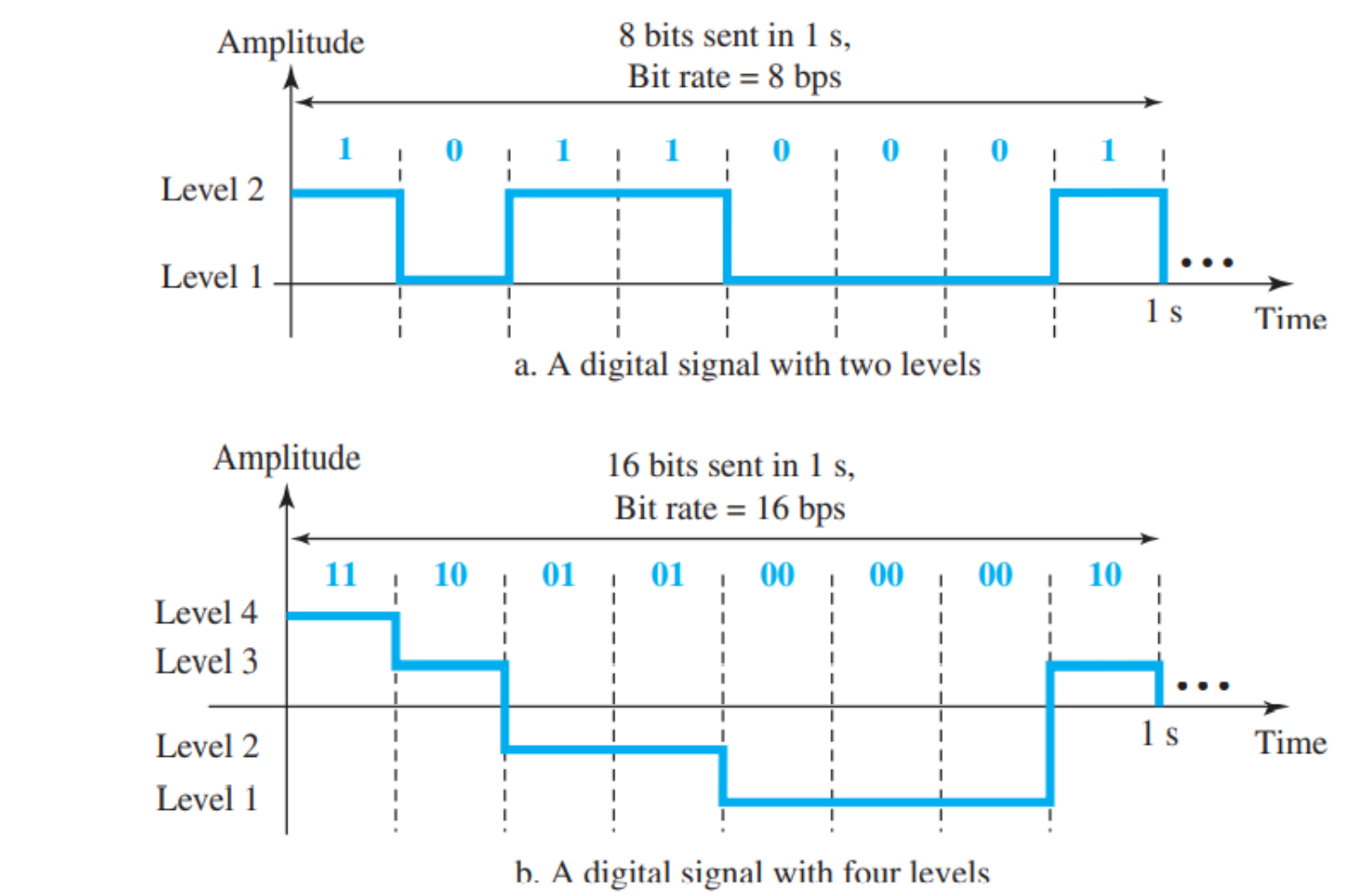
Contd...

- In FM, the instantaneous change in the carrier frequency is proportional to the amplitude of the modulating signal;
- In PM the instantaneous change in the carrier frequency is proportional to the derivative of the amplitude of the modulating signal.

Contd...


Amplitude
Modulating signal (audio)
Time
Carrier frequency
Time
PM signal
Time
VCO
d/dt
$B_{PM} = 2(1 + b)B$
0
f_c

DIGITAL SIGNALS



Problem – Concept Note

- A digital signal has eight levels. How many bits are needed per level?

- A digital signal has nine levels. How many bits are needed per level?

Bit Rate

- Bit rate (instead of frequency)—is used to describe digital signals.
- The bit rate is the number of bits sent in 1s, expressed in bits per second (bps).
- Assume we need to download text documents at the rate of 100 pages per second. What is the required bit rate of the channel? A page is an average of 24 lines with 80 characters in each line

Solution: 100 × 24 × 80 × 8 = 1,536,000 bps = 1.536 Mbps

Contd...

- A digitized voice channel is made by digitizing a 4-kHz bandwidth analog voice signal. We need to sample the signal at twice the highest frequency (two samples per hertz). We assume that each sample requires 8 bits. What is the required bit rate?

Solution: $2 \times 4000 \times 8 = 64{,}000$ bps $= 64$ kbps

Basic definitions

Bit Length

- The bit length is the distance one bit occupies on the transmission medium.

Bit length = propagation speed × bit duration

Throughput

The throughput is a measure of how fast we can actually send data through a network

1. A network with bandwidth of 10 Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of this network?

Throughput = (12,000 × 10,000) / 60 = 2 Mbps

Contd...

Latency (Delay)

The latency or delay defines how long it takes for an entire message to completely arrive at the destination from the time the first bit is sent out from the source.

Latency = propagation time + transmission time + queuing time + processing delay

Propagation time measures the time required for a bit to travel from the source to the destination.

Propagation time = Distance / (Propagation Speed)

The propagation speed of electromagnetic signals depends on the medium and on the frequency of the signal.

Contd...

Transmission Time The transmission time of a message depends on the size of the message and the bandwidth of the channel

$$\text{Transmission time} = (\text{Message size}) / \text{Bandwidth}$$

- The first bit may take a time equal to the propagation time to reach its destination; the last bit also may take the same amount of time.
- There is a time between the first bit leaving the sender and the last bit arriving at the receiver.
- The first bit leaves earlier and arrives earlier; the last bit leaves later and arrives later.

Contd...

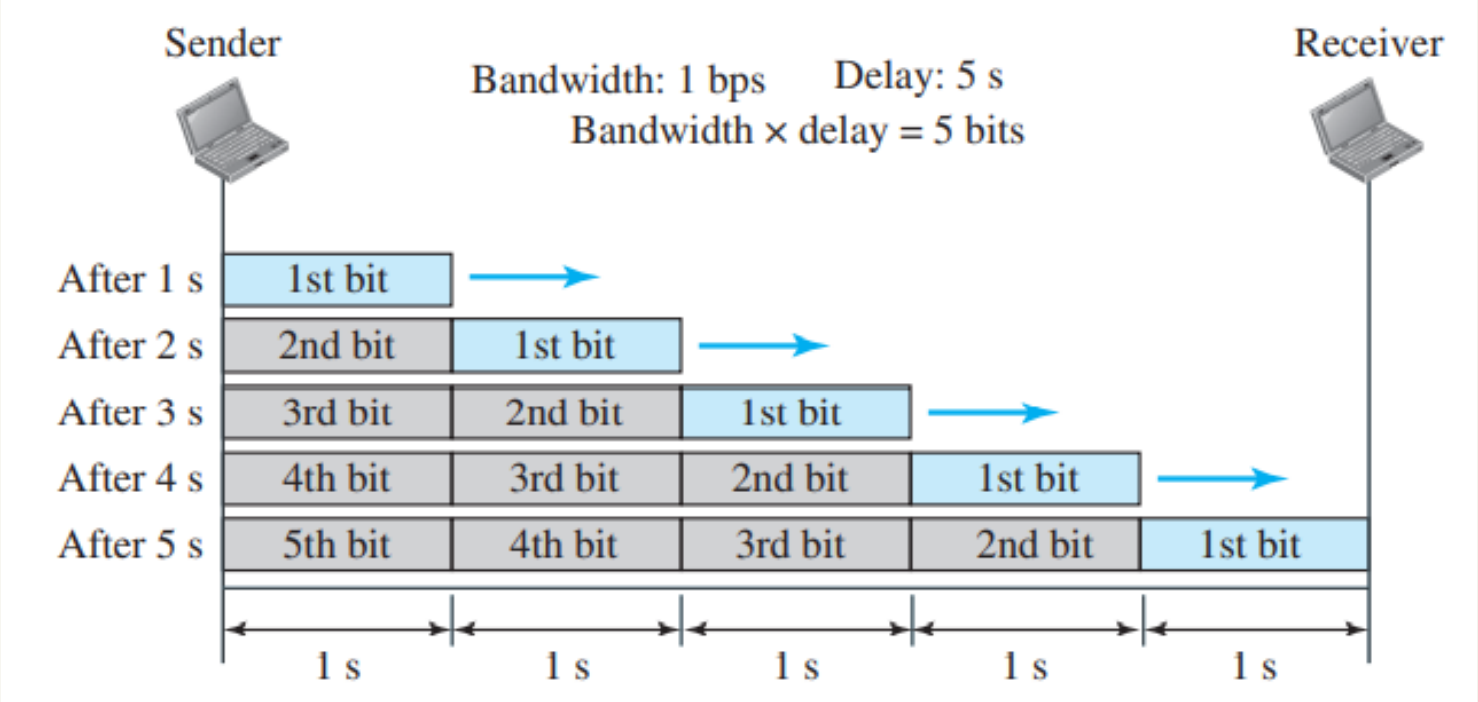
Queuing Time:

- The Third component in latency is the queuing time, the time needed for each intermediate or end device to hold the message before it can be processed.
- The queuing time is not a fixed factor; it changes with the load imposed on the network.
- When there is heavy traffic on the network, the queuing time increases.

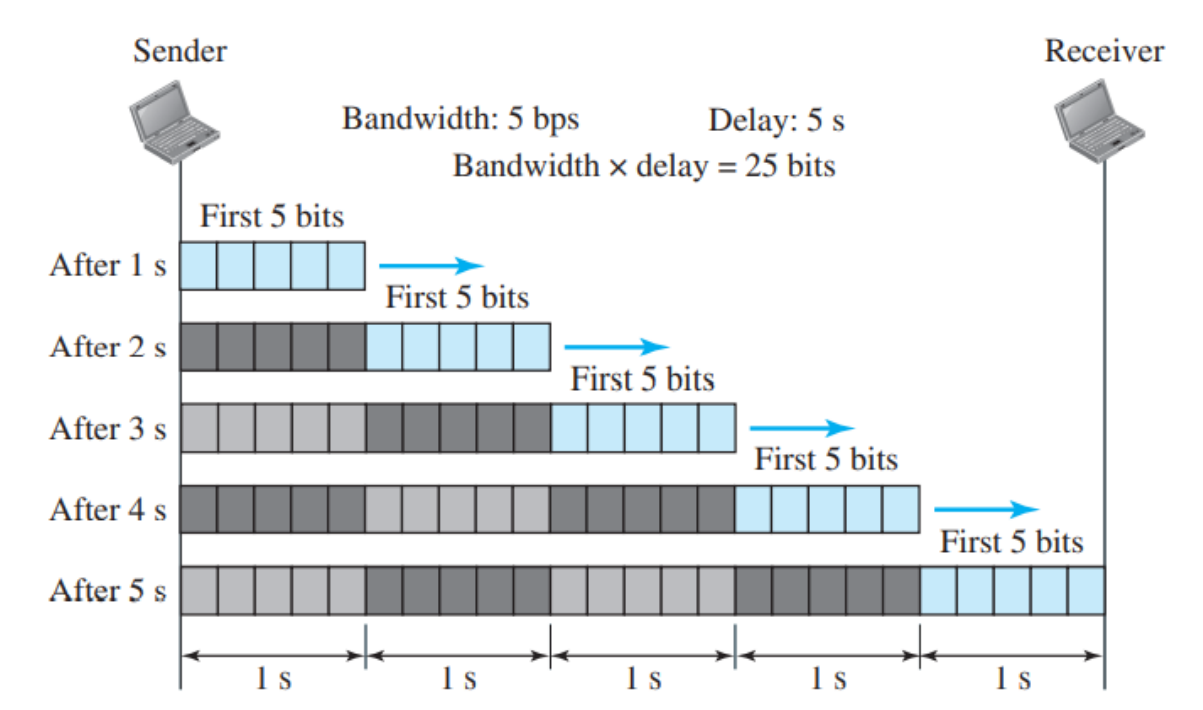
Basic definitions

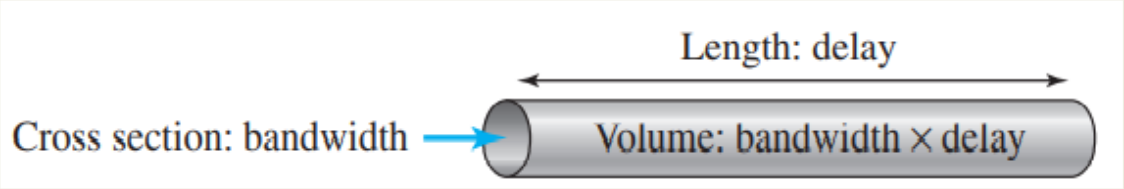
Bandwidth-Delay Product

Bandwidth and delay are two performance metrics of a link.



The bandwidth-delay product defines the number of bits that can fill the link.





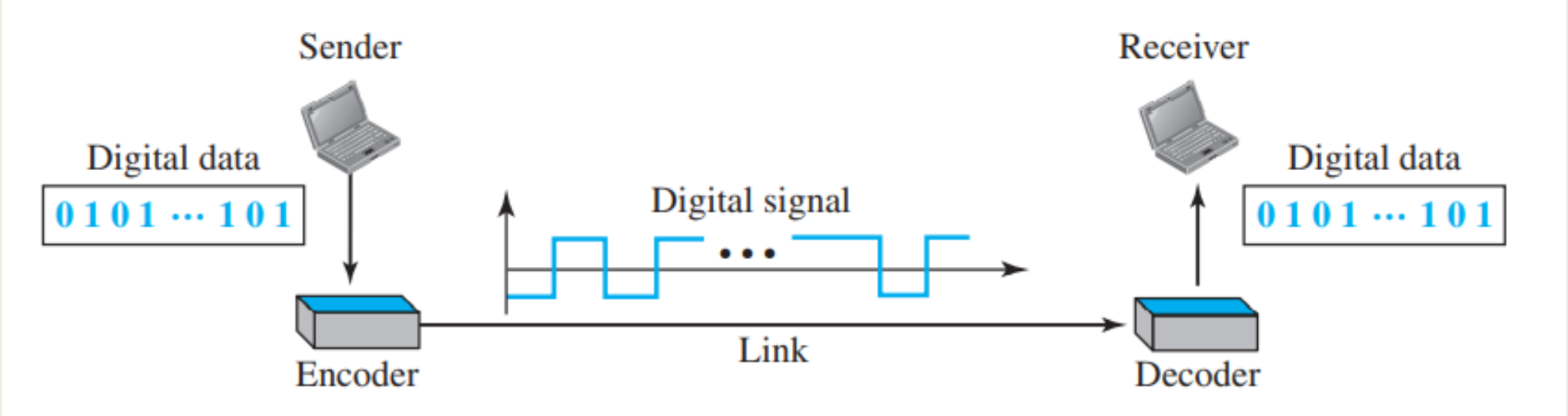
DIGITAL-TO-DIGITAL CONVERSION

- The conversion involves three techniques:
  - *line coding, block coding, and scrambling.*
- Line coding is always needed;
- Block coding and scrambling may or may not be needed.

Line Coding

- Line coding is the process of converting digital data to digital signals.
- The data, in the form of text, numbers, graphical images, audio, or video, are stored in computer memory as sequences of bits.
- Line coding converts a sequence of bits to a digital signal.
- At the sender, digital data are encoded into a digital signal; at the receiver, the digital data are recreated by decoding the digital signal.

Line coding and decoding



Signal Element Versus Data Element

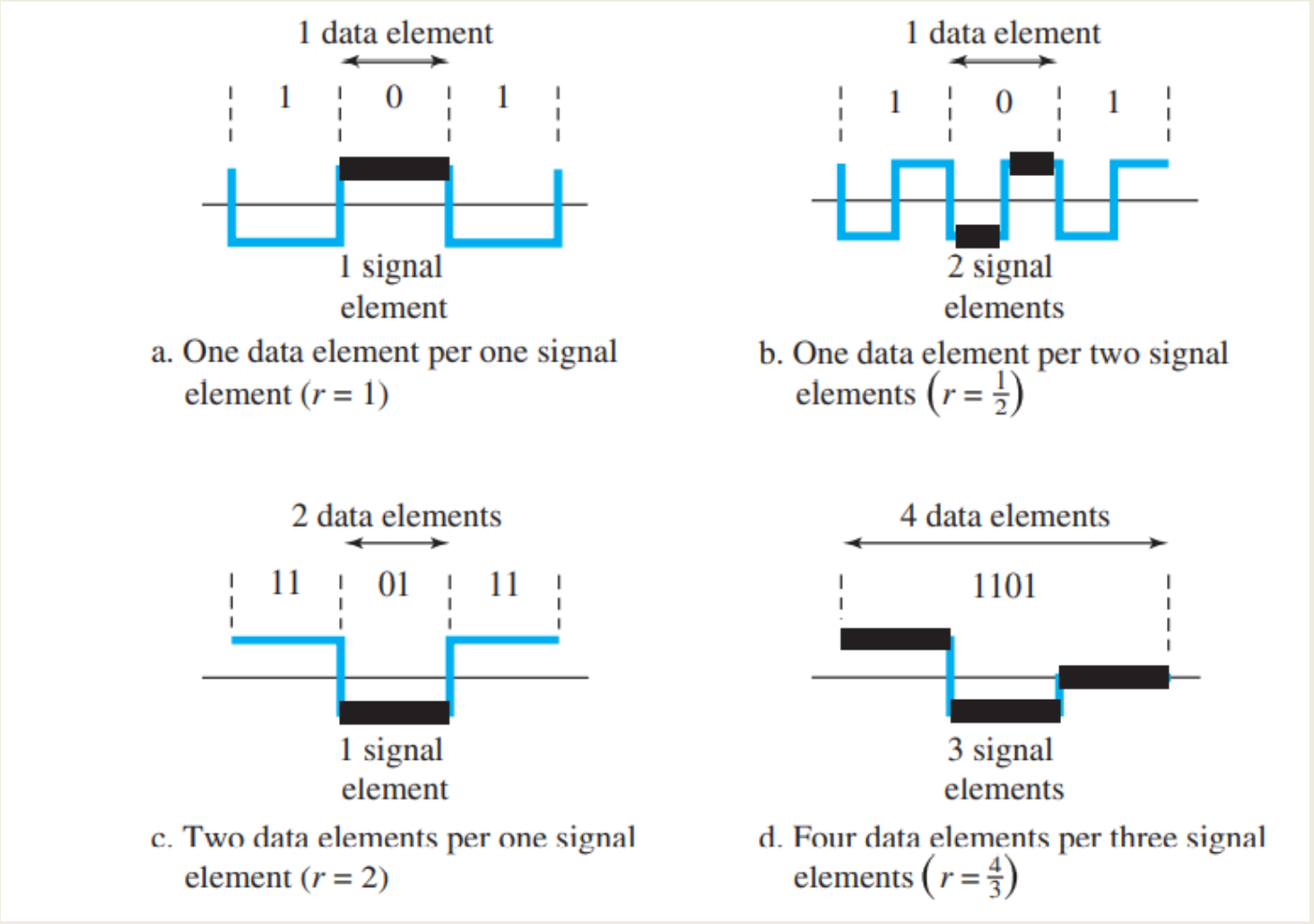
- In data communications, our goal is to send data elements.
- A ***data element*** is the smallest entity that can represent a piece of information: this is the bit.

1 0



- In digital data communications, a signal element carries data elements. A ***signal element*** is the shortest unit (timewise) of a digital signal.
- Data elements are being carried; signal elements are the carriers.

Contd...


1 data element
1
0
1
1 signal element
a. One data element per one signal element (r = 1)
1 data element
1
0
1
2 signal elements
b. One data element per two signal elements (r = 1/2)
2 data elements
11
01
11
1 signal element
c. Two data elements per one signal element (r = 2)
4 data elements
1101
3 signal elements
d. Four data elements per three signal elements (r = 4/3)

Data Rate Versus Signal Rate

- The ***data rate*** defines the number of data elements (bits) sent in 1s. The unit is bits per second (bps).
- The ***signal rate*** is the number of signal elements sent in 1s. The unit is the baud.
- The data rate is sometimes called the bit rate; the signal rate is sometimes called the pulse rate, the modulation rate, or the baud rate

Contd...

- One goal in data communications is to increase the data rate while decreasing the signal rate.
- Increasing the data rate increases the speed of transmission; decreasing the signal rate decreases the bandwidth requirement.
- The relationship between data rate (N) and signal rate (S)

$$S = N/r$$

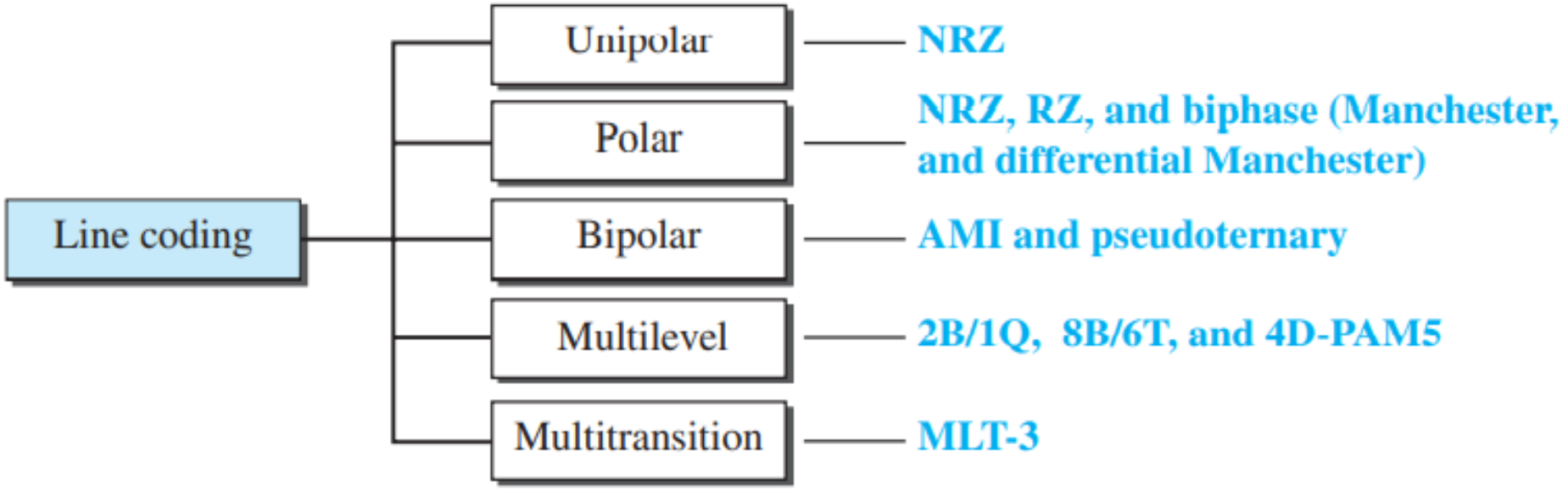
Contd...

- The relationship between data rate and signal rate as

$$S_{ave} = c \times N \times (1/r) \text{ baud}$$

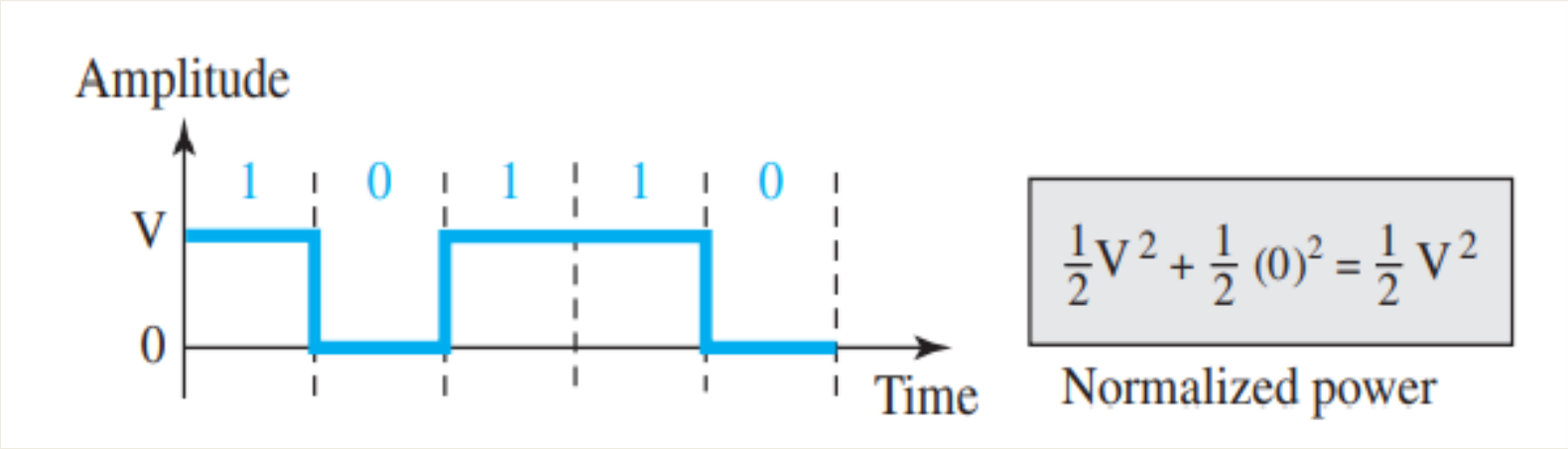
- where N is the data rate (bps); c is the case factor, which varies for each case; S is the number of signal elements per second.

Line Coding Schemes



Contd...

- Unipolar Scheme: In a unipolar scheme, all the signal levels are on one side of the time axis.



Contd...

- **NRZ (Non-Return-to-Zero):** a unipolar scheme was designed as a non-return-to-zero (NRZ) scheme in which the positive voltage defines bit 1 and the zero voltage defines bit 0.
- It is called NRZ because the signal does not return to zero at the middle of the bit.

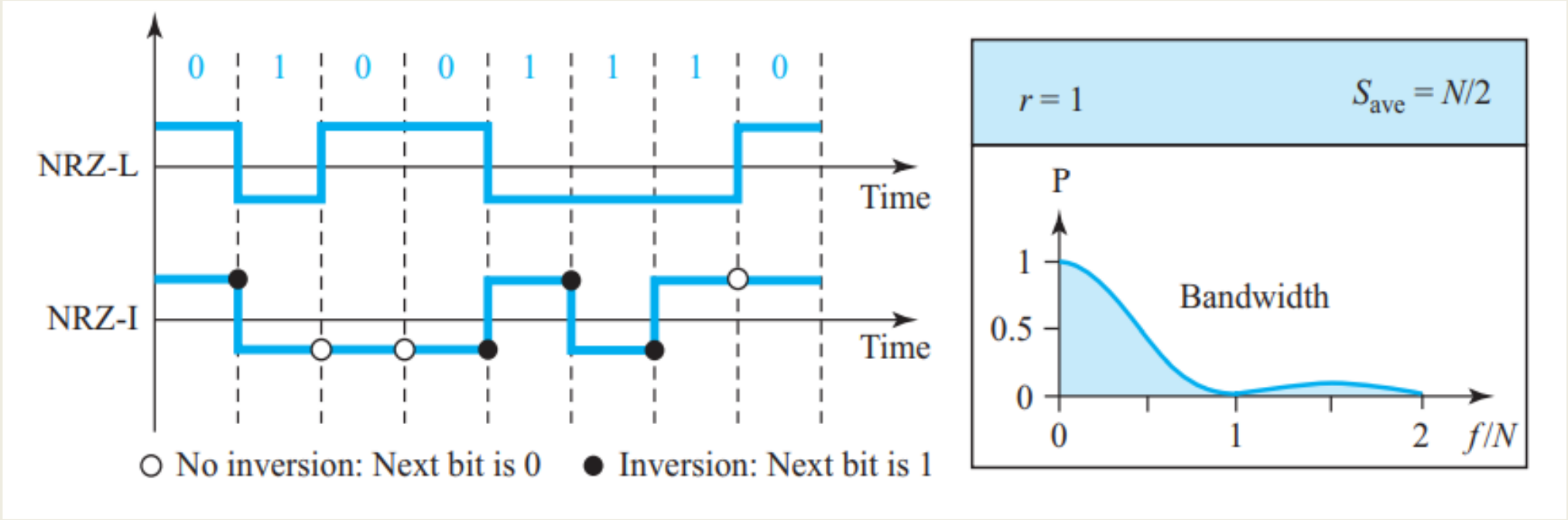
Polar Schemes

- In polar schemes, the voltages are on both sides of the time axis. For example,
  - *The voltage level for 0 can be positive*
  - *The voltage level for 1 can be negative*

Non-Return-to-Zero (NRZ)

- In polar NRZ encoding, we use two levels of voltage amplitude.
- **NRZ-L (NRZ-Level),** the level of the voltage determines the value of the bit.
- **NRZ-I (NRZ-Invert),** the change or lack of change in the level of the voltage determines the value of the bit. If there is no change, the bit is 0; if there is a change, the bit is 1.

Polar NRZ-L and NRZ-I schemes



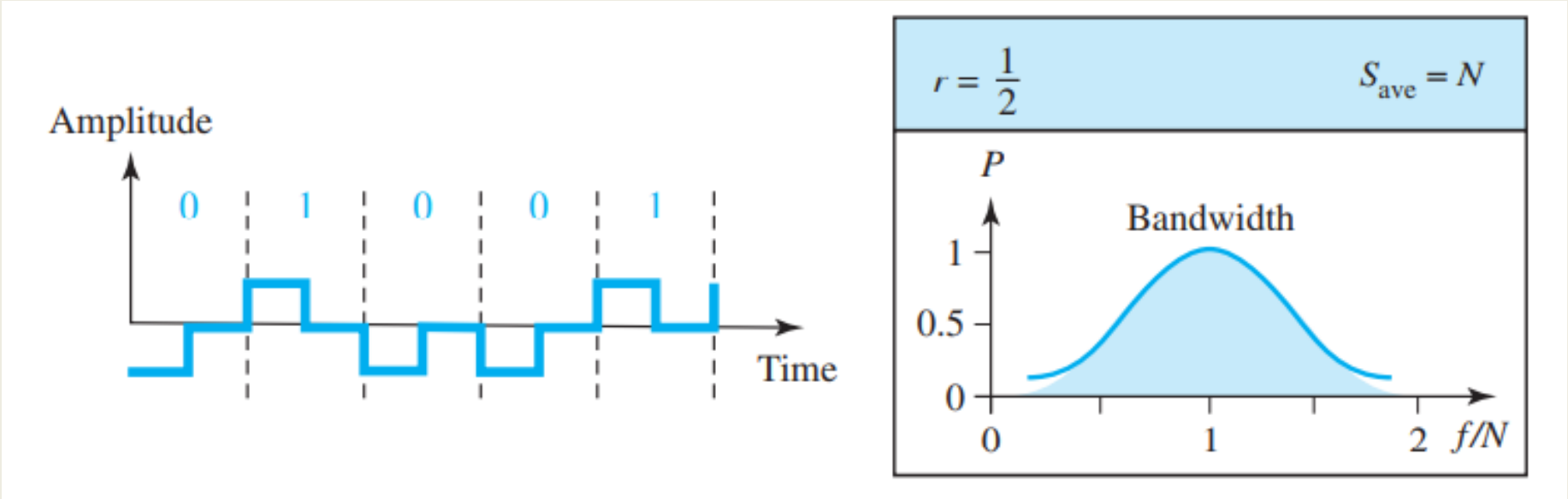
In NRZ-L the level of the voltage determines the value of the bit.

In NRZ-I the inversion or the lack of inversion determines the value of the bit.

Return-to-Zero (RZ)

- The main problem with NRZ encoding occurs when the sender and receiver clocks are not synchronized. The receiver does not know when one bit has ended and the next bit is starting.
- The return-to-zero (RZ) scheme, which uses three values: positive, negative, and zero.
- In RZ, the signal changes not between bits but during the bit.
- The main ***disadvantage of RZ*** encoding - it requires two signal changes to encode a bit and occupies **greater bandwidth**.

Contd...



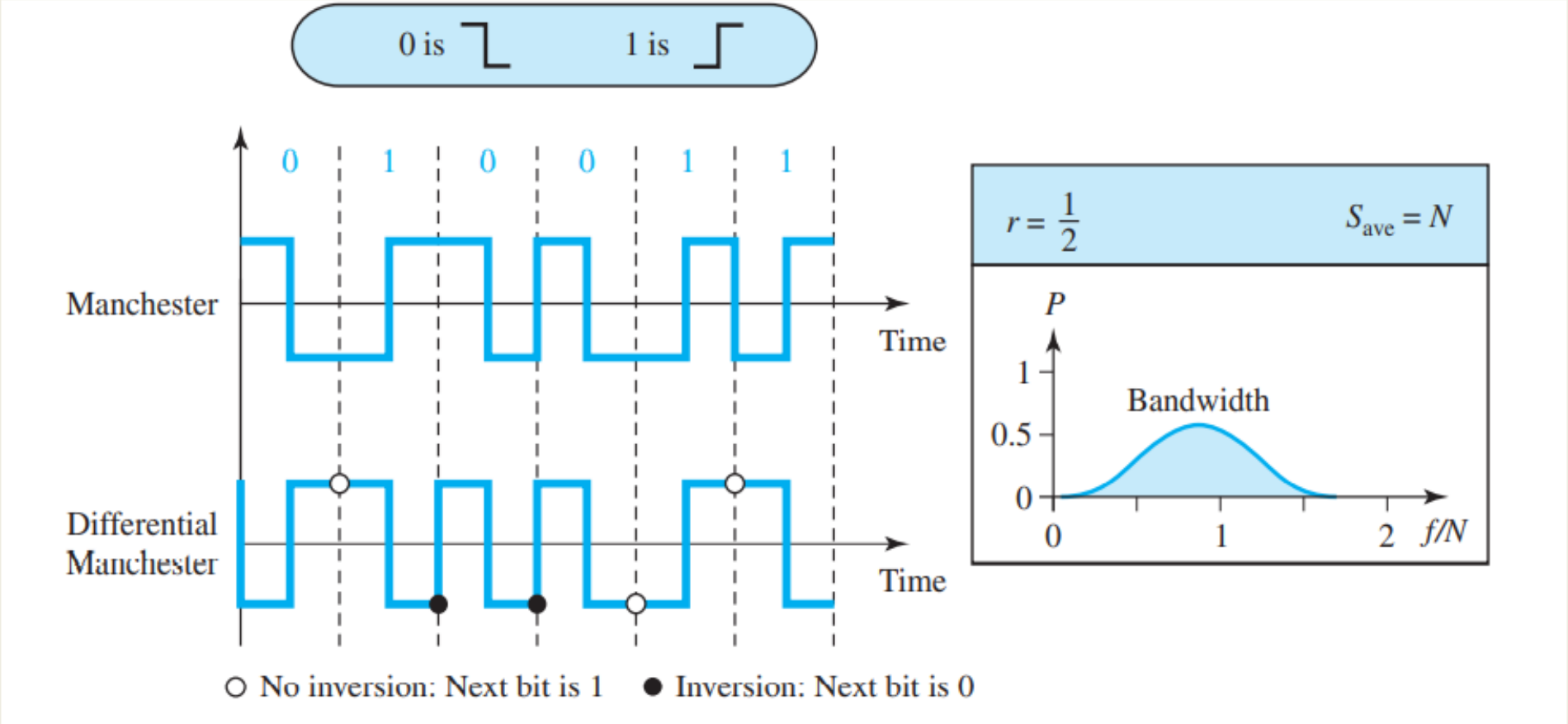
Contd...

- Biphase: Manchester and Differential Manchester
- The idea of RZ (transition at the middle of the bit) and the idea of NRZ-L are combined into the Manchester scheme.
- In Manchester encoding, the duration of the bit is divided into two halves.
- The voltage remains at one level during the first half and moves to the other level in the second half.

Contd...

- The transition at the middle of the bit provides synchronization. Differential Manchester, on the other hand, combines the ideas of RZ and NRZ-I.
- There is always a transition at the middle of the bit, but the bit values are determined at the beginning of the bit.
- If the next bit is 0, there is a transition; if the next bit is 1, there is none.

Polar biphase: Manchester and differential Manchester schemes



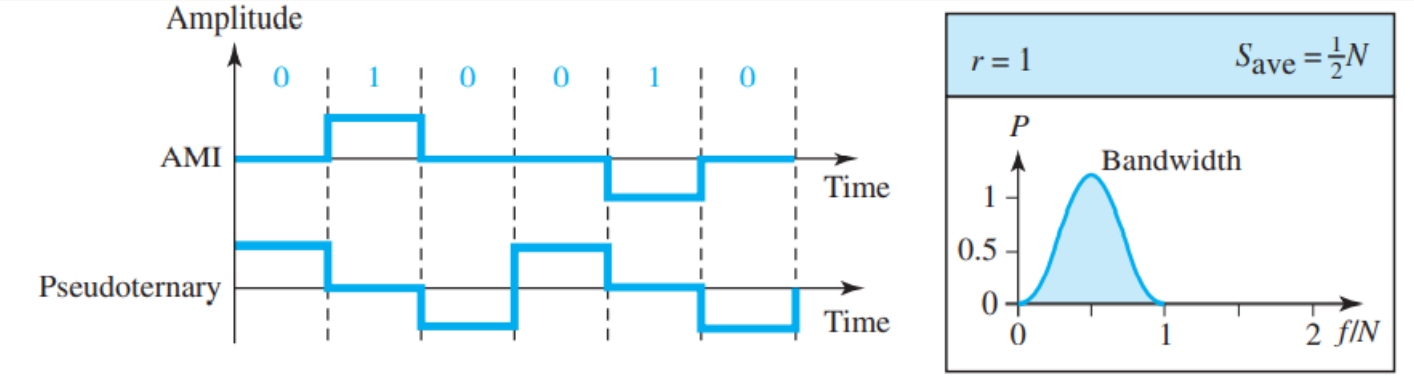
Contd...

- In Manchester and differential Manchester encoding, the transition at the middle of the bit is used for synchronization.
- The minimum bandwidth of Manchester and differential Manchester is 2 times that of NRZ

Bipolar Schemes

- In bipolar encoding (sometimes called multilevel binary), there are three voltage levels: positive, negative, and zero.
- The voltage level for one data element is at zero, while the voltage level for the other element alternates between positive and negative.
- AMI and Pseudoternary: Two variations of bipolar encoding: AMI and pseudoternary. A common bipolar encoding scheme is called bipolar alternate mark inversion (AMI).
- In the term alternate mark inversion, the word mark comes from telegraphy and means 1.So AMI means alternate 1 inversion.

- A variation of AMI encoding is called pseudoternary in which the 1 bit is encoded as a zero voltage and the 0 bit is encoded as alternating positive and negative voltages.
- Bipolar schemes: AMI and pseudoternary



Contd...

- The bipolar scheme was developed as an alternative to NRZ.
- The bipolar scheme has the same signal rate as NRZ, but there is no DC component.
- The NRZ scheme has most of its energy concentrated near zero frequency, which makes it unsuitable for transmission over channels with poor performance around this frequency.
- The concentration of the energy in bipolar encoding is around frequency N/2.

Multilevel Schemes

- The desire to increase the data rate or decrease the required bandwidth has resulted in the creation of many schemes.
- The goal is to increase the number of bits per baud by encoding a pattern of m data elements into a pattern of n signal elements.
- Two types of data elements (0s and 1s), which means that a group of m data elements can produce a combination of 2m data patterns.

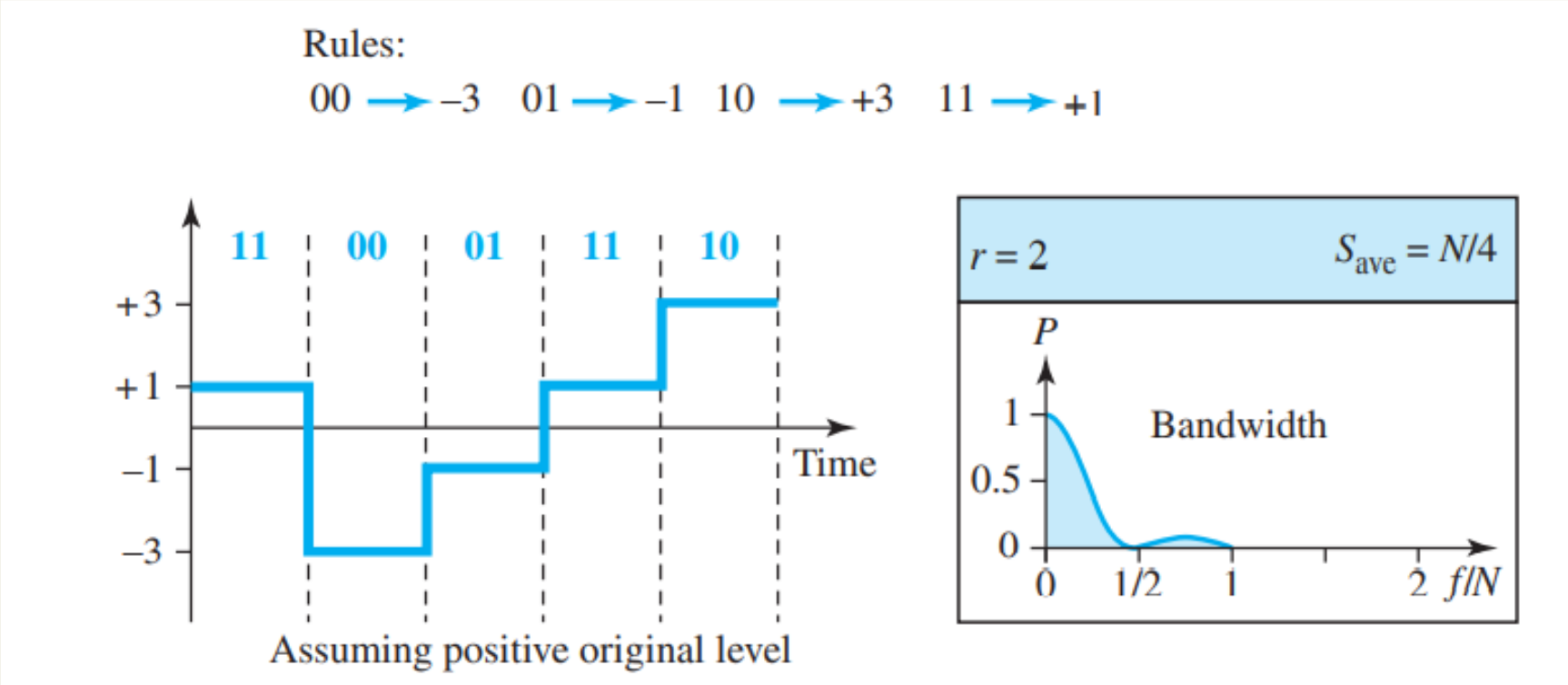
Contd...

- The code designers have classified these types of coding as mBnL, where
  - *m is the length of the binary pattern*
  - *B means binary data*
  - *n is the length of the signal pattern*
  - *L is the number of levels in the signaling.*
- In mBnL schemes, a pattern of m data elements is encoded as a pattern of n signal elements in which $2^m \leq L^n$.

mBnL

- 2B1Q The first mBnL scheme, two binary, one quaternary (2B1Q), uses data patterns of size 2 and encodes the 2-bit patterns as one signal element belonging to a four-level signal.
- In this type of encoding m = 2, n = 1, and L = 4 (quaternary).

Multilevel: 2B1Q scheme



Contd...

- The average signal rate of 2B1Q is $S = N/4$.
- 2B1Q, we can send data 2 times faster than by using NRZ-L.
- However, 2B1Q uses four different signal levels, which means the receiver has to discern four different thresholds.
- The reduced bandwidth comes with a price.
- There are no redundant signal patterns in this scheme because $2^2 = 4^1$.
- The 2B1Q scheme is used in DSL (Digital Subscriber Line) technology to provide a high-speed connection to the Internet by using subscriber telephone lines.

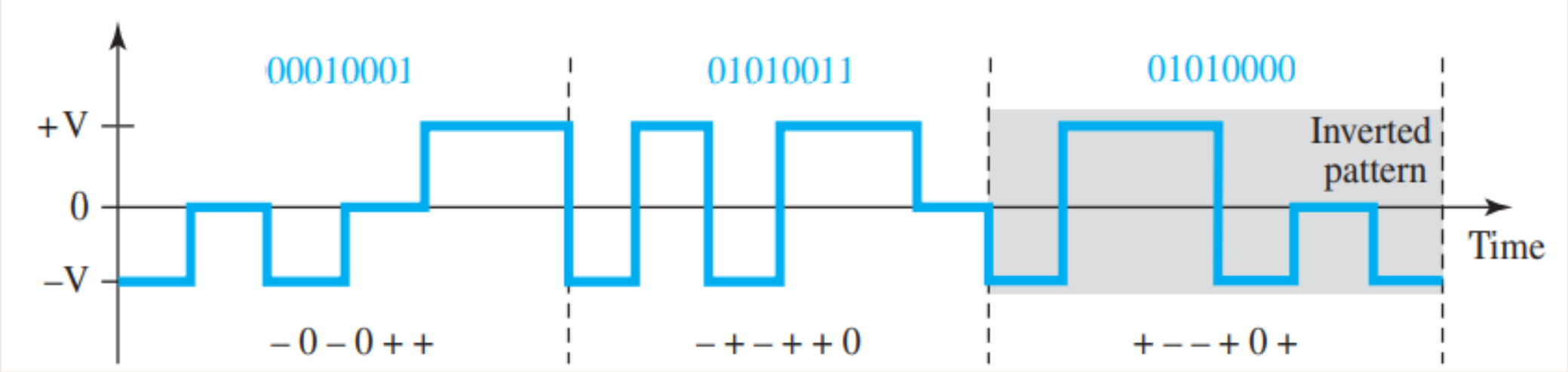
8B6T

- A very interesting scheme is eight binary, six ternary (8B6T).
- This code is used with 100BASE-4T cable
- The idea is to encode a pattern of 8 bits as a pattern of six signal elements, where the signal has three levels (ternary).
- In this type of scheme, we can have $2^8 = 256$ different data patterns and $3^6 = 729$ different signal patterns.
- There are $729 - 256 = 473$ redundant signal elements that provide synchronization and error detection

Contd...

- The three possible signal levels are represented as −, 0, and +.
- The first 8-bit pattern 00010001 is encoded as the signal pattern − 0 − 0 + + with weight 0;
- The second 8-bit pattern 01010011 is encoded as − + − + + 0 with weight +1.
- The third 8-bit pattern 01010000 should be encoded as + − − + 0 + with weight +1
- To create DC balance, the sender inverts the actual signal. The receiver can easily recognize that this is an inverted pattern because the weight is −1. The pattern is inverted before decoding.

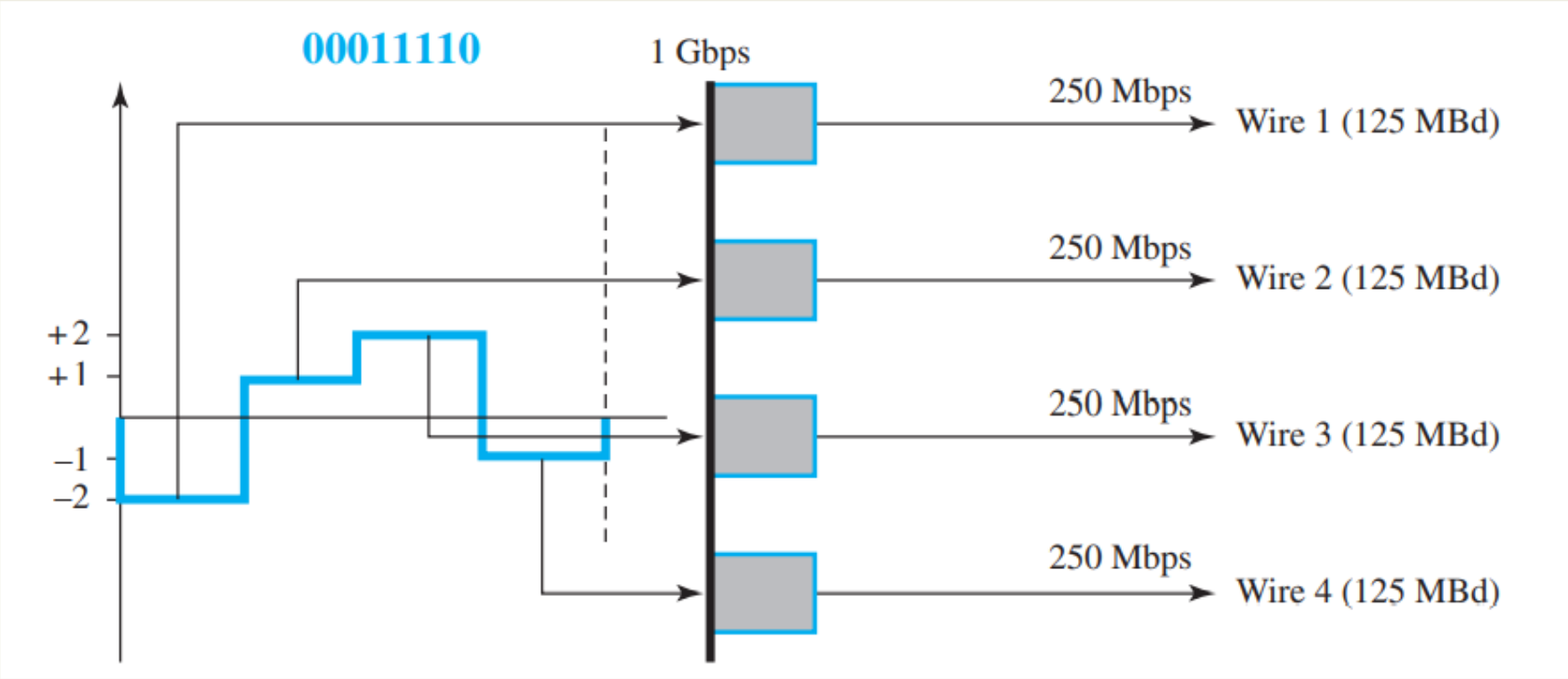
Contd...


00010001
01010011
01010000
+V
0
–V
Inverted
pattern
Time
– 0 – 0 + +
– + – + + 0
+ – – + 0 +

4D-PAM5

- Four-dimensional five level pulse amplitude modulation (4D-PAM5). The 4D means that data is sent over four wires at the same time.
- It uses five voltage levels, such as −2, −1, 0, 1, and 2.
- one level, level 0, is used only for forward error detection

Contd...


00011110
1 Gbps
+2
+1
–1
–2
250 Mbps
Wire 1 (125 MBd)
250 Mbps
Wire 2 (125 MBd)
250 Mbps
Wire 3 (125 MBd)
250 Mbps
Wire 4 (125 MBd)

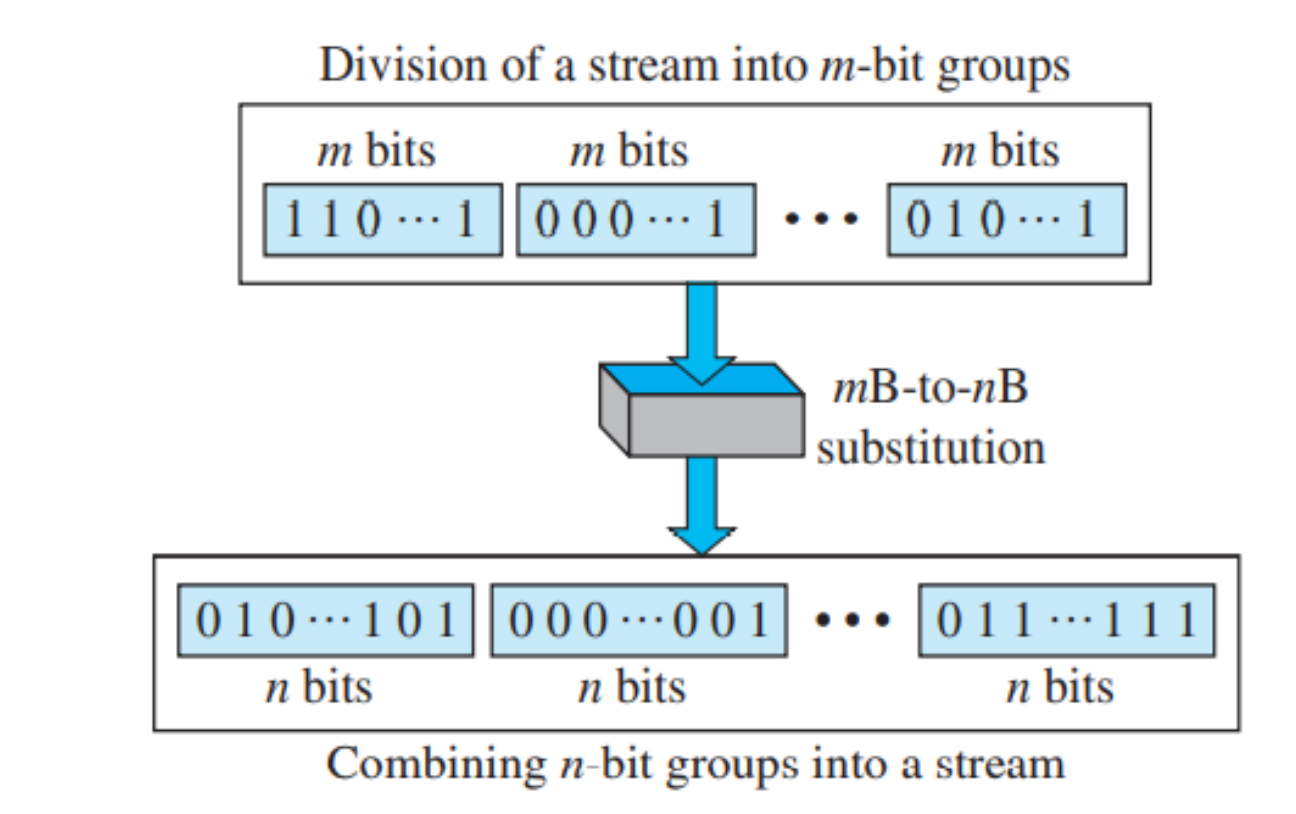
Block Coding

- In general, block coding changes a block of m bits into a block of n bits, where n is larger than m.
- Block coding is referred to as an mB/nB encoding technique.
- The slash in block encoding (for example, 4B/5B) distinguishes block encoding from multilevel encoding (for example, 8B6T)
- Block coding normally involves three steps: division, substitution, and combination.

Contd...

- In the division step, a sequence of bits is divided into groups of m bits. For example, in 4B/5B encoding, the original bit sequence is divided into 4-bit groups.
- The heart of block coding is the substitution step. In this step, we substitute an m-bit group with an n-bit group.
- For example, in 4B/5B encoding we substitute a 4-bit group with a 5-bit group.
- Finally, the n-bit groups are combined to form a stream. The new stream has more bits than the original bits

Block coding concept



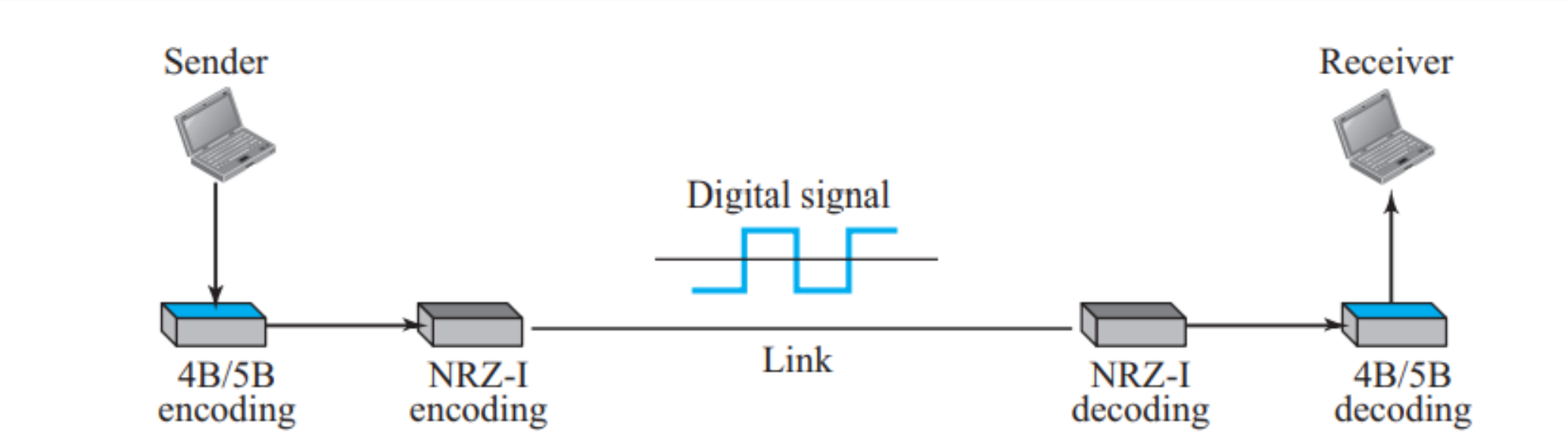
4B/5B

- The four binary/five binary (4B/5B) coding scheme was designed to be used in combination with NRZ-I.
- NRZ-I has a good signal rate, one-half that of the biphase, but it has a synchronization problem.
- A long sequence of 0s can make the receiver clock lose synchronization.
- One solution is to change the bit stream, prior to encoding with NRZ-I, so that it does not have a long stream of 0s. The 4B/5B scheme achieves this goal.

Contd...

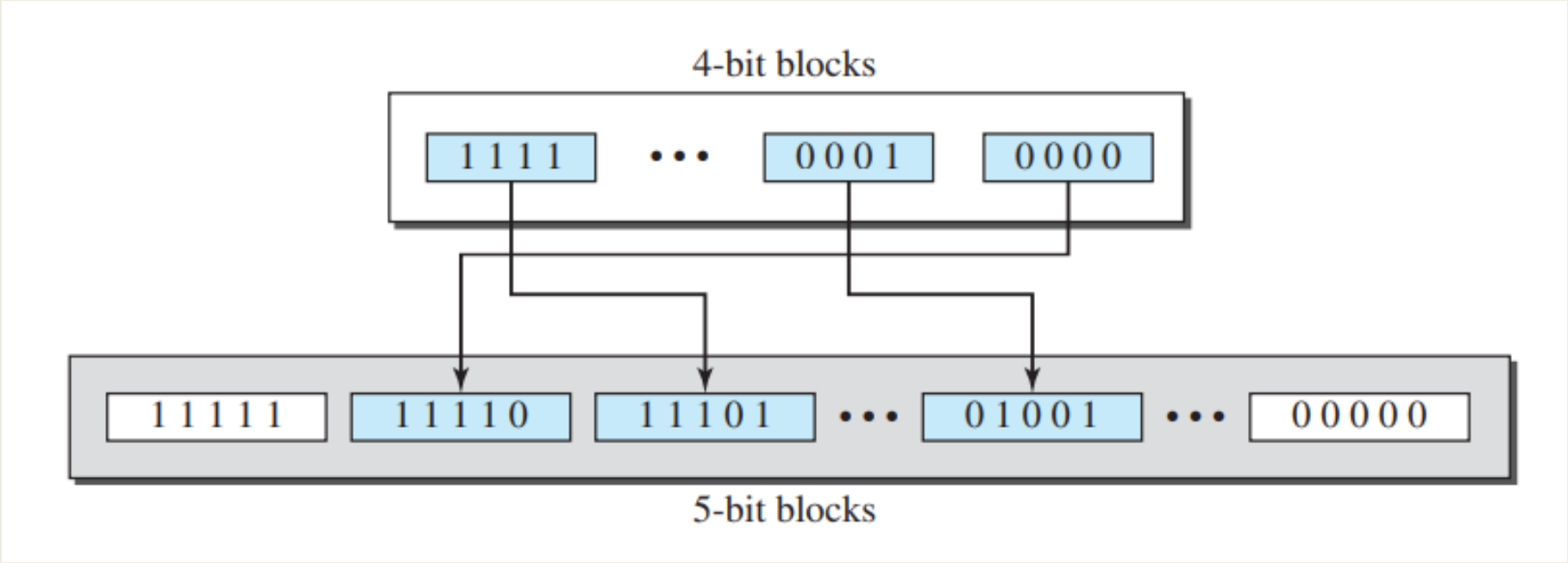
- The block-coded stream does not have more that three consecutive 0s.
- At the receiver, the NRZ-I encoded digital signal is first decoded into a stream of bits and then decoded to remove the redundancy.
- In 4B/5B, the 5-bit output that replaces the 4-bit input has no more than one leading zero (left bit) and no more than two trailing zeros (right bits).
- when different groups are combined to make a new sequence, there are never more than three consecutive 0s.

Using block coding 4B/5B with NRZ-I line coding scheme



Data Sequence	*Encoded Sequence*	*Control Sequence*	*Encoded Sequence*
0000	11110	Q (Quiet)	00000
0001	01001	I (Idle)	11111
0010	10100	H (Halt)	00100
0011	10101	J (Start delimiter)	11000
0100	01010	K (Start delimiter)	10001
0101	01011	T (End delimiter)	01101
0110	01110	S (Set)	11001
0111	01111	R (Reset)	00111
1000	10010		
1001	10011		
1010	10110		
1011	10111		
1100	11010		
1101	11011		
1110	11100		
1111	11101		

Substitution in 4B/5B block coding



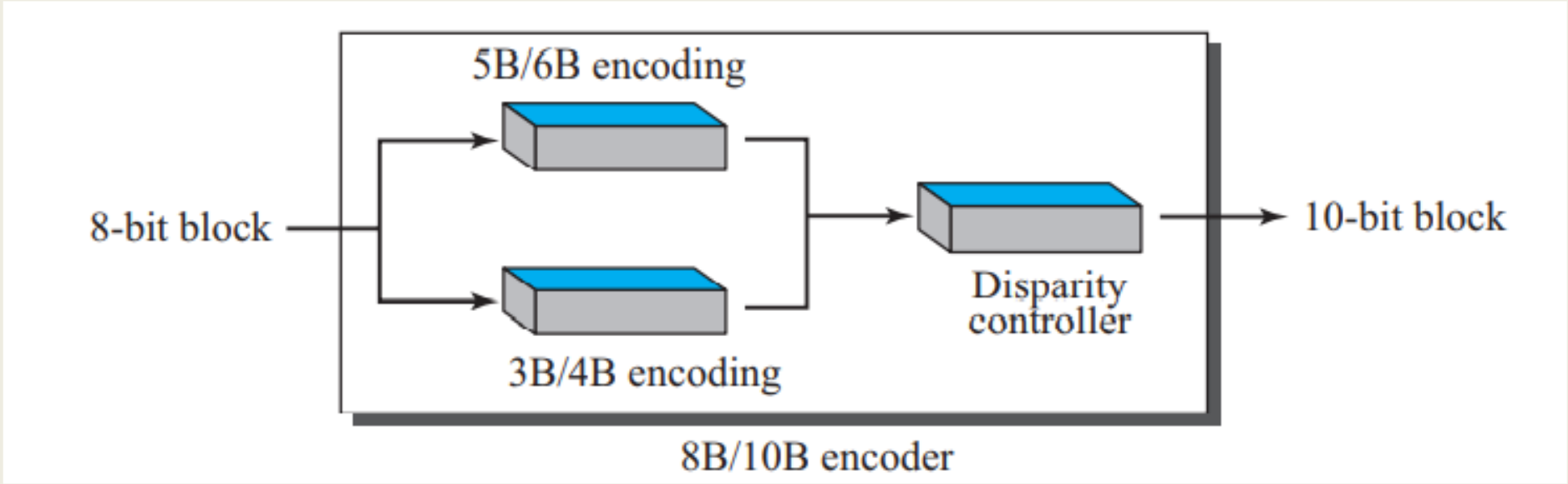
8B/10B

- The eight binary/ten binary (8B/10B) encoding is similar to 4B/5B encoding except that a group of 8 bits of data is now substituted by a 10-bit code.
- It provides greater error detection capability than 4B/5B.
- The 8B/10B block coding is actually a combination of 5B/6B and 3B/4B encoding.
- The five most significant bits of a 10-bit block are fed into the 5B/6B encoder; the three least significant bits are fed into a 3B/4B encoder.
- The split is done to simplify the mapping table.

Contd...

- To prevent a long run of consecutive 0s or 1s, the code uses a disparity controller which keeps track of excess 0s over 1s (or 1s over 0s).
- If the bits in the current block create a disparity that contributes to the previous disparity (either direction), then each bit in the code is complemented (a 0 is changed to a 1 and a 1 is changed to a 0).
- The coding has $210 - 28 = 768$ redundant groups that can be used for disparity checking and error detection.
- In general, the technique is superior to 4B/5B because of better built-in error-checking capability and better synchronization.

8B/10B block encoding



Exit Clips

- 11001010 – NRZ, RZ, NRZ-L, NRZ – I and Manchester coding

Thank You